

Do You See What I See? Evaluating Relative Depth Judgments Between Real and Virtual Projections

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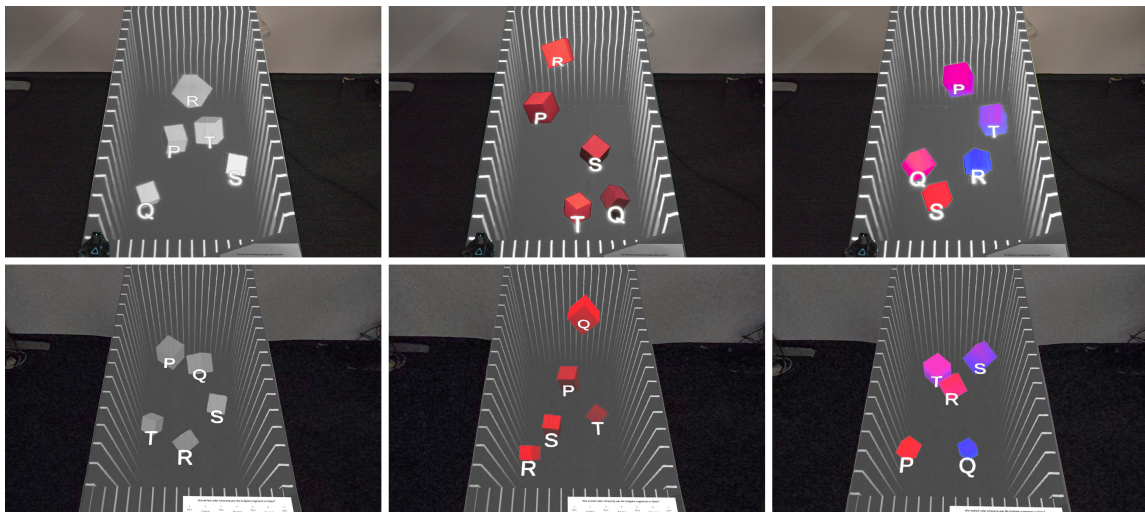


Figure 1: Visualization concepts rendered using a physical projector-based AR approach (top row) and a virtual simulation using video see-through AR (bottom row). Visualization concepts from left to right: *Shaded, Fog, Pseudo-Chromadepth*.

Abstract

Projector-based augmented reality (AR) is promising in different domains with less issues in discomfort or shortage of space. However, due to limitations like high costs and cumbersome calibration, this AR modality remains underused. To address this problem, a stereoscopic projector-based AR simulation was implemented for a cost-effective video see-through AR headset. To evaluate the validity

of this simulation, a relative depth judgment experiment was conducted to compare this method with a physical projection system. Consistent results suggest that a known interaction effect between visualization and disparity mode could be successfully reproduced using both the physical projection and the virtual simulation. In addition, first findings indicate that there are no significant differences between these projection modalities. The results indicate that other perception-related effects observed for projector-based AR may also be applicable to virtual projection simulations and that future findings determined using only these simulations may also be applicable to real projections.

CCS Concepts

• **Computing methodologies** → **Mixed / augmented reality; Perception**; • **Hardware** → **Displays and imagers**; • **Human-centered computing** → **Empirical studies in visualization**.

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Keywords

Projector-based Augmented Reality, Depth Perception, Simulation

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1 Introduction

Projector-based augmented reality (AR), also known as spatial AR, is a method that extends physical objects with virtual contents using projectors. This can be used to either change visual surface properties, or to render spatially aligned 3D objects [4]. Because of the spatial separation of the projectors from the visualization, there are fewer issues in terms of sterility, discomfort and shortage of space, when compared to alternative AR displays. Moreover, cognitive cost and performance could be shown to be improved by this method [2, 19]. Common application areas are, e.g., medical scenarios [15] or assembly guidance [9]. The acquisition, setup, calibration, and spatial registration of such systems are relatively costly, time-intensive, and prone to errors. In addition, due to the complex calibration and cumbersome setup, projector-based AR is often installed in a fixed (indoor) location. This has limited the adoption of the technique in potential useful application areas, despite its demonstrated advantages. Additionally, tracking and registration inaccuracies create uncertainties that can negatively impact usability assessments of AR content. Therefore, research on projector-based AR is also relatively sparse compared to other, more affordable, approaches [39].

In this context, real-world studies have increasingly been recreated in other extended reality (XR) modalities, including virtual reality (VR) [32, 35, 40, 53, 54]. This can provide controlled and standardized evaluation environments in different domains [22, 29, 42]. Examples include medical visualizations [20, 39], interaction techniques [43], medical robotics [38], AR visualization in logistics [44], airborne surveillance [1], and industrial maintenance [7]. Empirical research in XR simulations offers distinct advantages. VR enables controlled evaluations of AR visualizations with consistent object geometry and reduced external influences like tracking inaccuracies [33, 34]. It also saves time and costs while allowing earlier, efficient evaluations [35], also using methods of rapid prototyping [16, 36]. XR can simulate AR devices for robust and reproducible testing [26], as demonstrated for AR displays in navigation [23] and hand-held AR systems [8, 55]. For optical see-through HMDs, VR simulations closely mirror real AR applications [26].

Therefore, it appears promising to also use alternative XR displays to more easily evaluate projector-based AR systems as this could mitigate identified issues with projection systems. This was already shown to be feasible with monoscopic 2D visualizations in a medical navigation context [39]. However, it remains open how well projections of more spatial or stereoscopic visualizations can be simulated. Especially perception-related properties are of interest here, since depth judgments are known to be a major issue in AR [25, 52]. Projector-based AR faces additional unique challenges

due to its distinct display properties, e.g., distortions, reduced resolution, and shadowing effects. These properties can conflict with depth cues from physical surfaces [5]. Studies have shown that while monoscopic setups rely on cues like motion parallax and shading [5], stereoscopic projections provide more accurate depth judgments, especially for objects closer to the projection surface [6, 49]. Additionally, manipulations using pictorial and binocular cues can effectively alter depth judgments [48, 50]. However, prior work also showed there is significant interaction between altering the visualization type of virtual objects and using binocular disparity, with stereopsis having less effect on visualizations that already convey sufficient depth information on their own [18]. In general, perception in AR and VR has been improved through various visualization techniques, ranging from occlusion-based methods like virtual windows [3] and transparency effects [51] to auxiliary visualizations such as glyph-based encodings [27, 31]. Shading-based approaches, including pseudo-chromadepth and aerial perspective, have consistently enhanced depth judgments [17, 24]. Illustrative techniques, such as supporting lines, hatching, and depth-encoded shading, can provide additional spatial cues [27, 28, 45]. Other methods, including random dot overlays [37] or cast shadows [10], may further refine depth perception depending on scene complexity. The choice of technique depends on application requirements, task demands, and the available depth cues within a given visualization.

This work aims to foster the dissemination and evaluation of projector-based AR by developing a simulation of a stereoscopic projection system. This simulation is visualized using latest developments in the field of video see-through (VST) AR head-mounted display (HMD). Since perceptual aspects of depth judgments are highly relevant in AR and have already been extensively researched, this work attempted to replicate a known experimental setup from related work to evaluate the validity of this simulation in terms of perception. A relative depth judgment task with varying visualization and binocular disparity conditions is performed using both a physical projector-based AR system and the developed simulation. Given the variety of depth-enhancing techniques, surface-based visualization methods were used in this work due to their frequent use in visualization tasks [41] and their established effectiveness in prior depth judgment experiments [18, 24, 28]. The overall objective of this study is to investigate whether similar effects can be observed in both modality conditions, i.e., physical and virtual projector-based AR.

2 Technical Methods

All methods were implemented using the engine *Unity* (Unity Technologies, USA), run on a computer with a 3.0 GHz quad core *Intel(R) Core(TM) i5-4590s* processor with 16 GB RAM and a *NVIDIA GeForce GTX 1080 Ti* graphics card.

2.1 Visualization Concepts

Previous work found significant differences between visual appearances of virtual objects for sorting tasks [18, 24, 28]. As the primary goal of this work was not to evaluate specific shading approaches, but to investigate differences between physical and virtual projection systems, three pertinent approaches from related work with known effects on depth judgments were selected. All concepts were

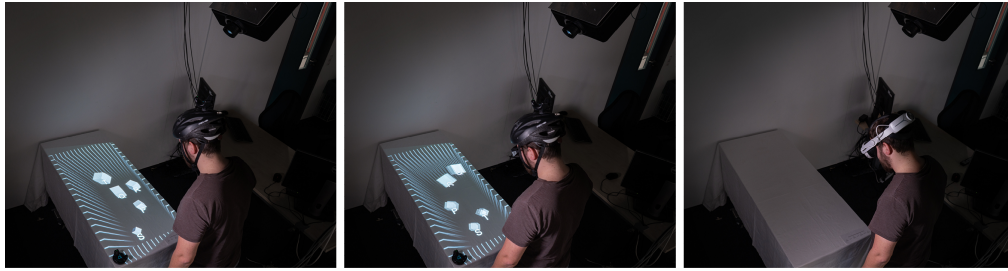


Figure 2: The study apparatus with user solving the sorting task. Left: *Mono Projection*, Middle: *Stereo Projection*, Right: *VST*.

implemented as Cg-based shaders and can be seen in Figure 1. As a first technique, a method based on Phong shading was selected. This *Shaded* condition is often used in surface visualization [41] and served as baseline in many other studies before [11, 18, 28]. Objects were opaquely rendered in a light gray color and illuminated by a directional light source positioned above them. In addition, two depth-enhancing surface visualization approaches were implemented. In the *Fog* condition, the color of objects was decreased in saturation and luminance depending on their distance to the viewer. This visualization was based on the concept of aerial perspective and was also investigated in depth judgment experiments before [17, 24]. The implemented shader was based on the *Shaded* condition, i.e., objects were also illuminated following a Phong approach. In addition, surface colors were selected based on the height of the vertices within a specified bounding volume using a color gradient texture. Colors ranged from bright red to a blueish dark red hue. The second depth-enhancing visualization method was pseudo-chromadepth (*PCD*) shading [46]. This approach was based on color temperature and has proven itself to enable superior depth judgments in prior experiments [18, 24, 48]. Flat shaded objects were rendered on a red-to-blue color scale depending on their height in a bounding volume. To this end, the implementation also used a color gradient texture.

2.2 Modality Factor

This section details the technical apparatus of this work’s physical and virtual projector-based AR implementations.

2.2.1 Physical Projector-based AR Environment. A Barco G60-W7 DLP laser phosphor projector (Barco GmbH, Germany) with a resolution of 1920x1080 px was diagonally mounted above a table serving as projection surface. Stereoscopic vision was enabled by setting the projector to side-by-side stereo and using 3D shutter glasses (Eyes 3 Shut, France). The shutter glasses were only used during stereo mode. The photogrammetric measurement system *ProjectionTools* [12] was used to calibrate the projector and to obtain a surface mesh of the projection area. Resulting data was then used to perform static projection mapping within Unity. This was implemented using the *domeprojection.com Unity Plugin* [13] and enabled virtual content positioned below the projection area to be rendered perspectively correctly from the viewer’s position. Head tracking was realized using an *HTC Vive Tracker* attached to a helmet. For monoscopic projections, an estimate of the eye position relative to the helmet marker was used as eye point and only a single image

was projected. For stereopsis, the scene was rendered twice, for the left and right eye individually. This setup is shown in Figure 2.

2.2.2 Virtual Projector-based AR Environment. A Meta Quest 3 (Meta Platforms, Inc., USA) was used in pass-through mode as a VST AR HMD to create a simulation of projector-based AR. The *Meta XR All-in-One SDK* was used together with the pass-through and spatial anchor building blocks to operate the device. The projector-based AR simulation was based on the *Virtual CAVE* demo scene of *Projector Simulator Pro* [14] Unity asset. The walls of the virtual CAVE were adjusted to the size of the physical projection surface and limited to that plane only. Position and settings of the virtual projector, e.g., aspect ratio and field of view, were adjusted to closely match the real projector. For monoscopic projections, only one virtual projector was used. The Meta camera rig center eye position was used for perspective calculations. The projection was then displayed on the virtual projection surface and rendered by the center eye of the Meta camera rig to convey it to the headset user. For stereoscopic projections, the rig was set to use per eye cameras and two virtual projectors were used to render the perspectively correct image for each individual eye position. Registration of the virtual projection surface to the real surface was done manually using the Quest 3 controllers. Once aligned, this position was maintained using spatial anchors. The virtual contents were placed below the projection surface and rendered perspectively correctly using the *Projector Simulator Pro* projection mapping.

3 Evaluation Methods

The primary objective of this work was to determine whether known perceptual effects of projector-based AR could be replicated using physical projections and a projector simulation displayed through a VST AR HMD. As an initial evaluation, it was decided to re-implement a relative depth judgment task, which was shown to be affected by an interaction effects between visualization method and binocular disparity [18]. Three factors were considered for this experiment. The first two factors were selected in accordance with the study conducted by Heinrich et al. [18]. Hence, the *Visualization* factor consisted of the implemented rendering methods and had the three respective levels *Shaded*, *Fog*, and *PCD*. The second factor manipulated the binocular *Disparity*. Its levels differentiated between monoscopic (*Mono*) and stereoscopic (*Stereo*) projections. The third factor examined the *Modality* with which the projected visualizations were displayed. This factor had the respective two

levels *Projection*, i.e., the real projector-based AR setup, and *VST*, i.e., the virtual projector-based AR setup visualized using the VST AR headset. In this within-subjects study, each subject tested each factor level combination resulting in a balanced $3 \times 2 \times 2$ design.

One half of the experiment could be considered a replication of the experiment reported by Heinrich et al. [18]. Hence, it was assumed that that study's respective findings can also be shown in this work and postulated the first hypothesis as: *H1 : There is a significant Visualization and Disparity main effect, as well as, interaction effect between these factors, across dependent measures.* The other half of the experiment consists of trials conducted using a simulation shown in a VST AR HMD. This work aimed at showing that there are no differences between depth judgments in real and in simulated projector-based AR. Thus, it was hypothesized that: *H2 : Across variables, there are no significant differences between Modality levels..* In addition, to also show that the "real" and "virtual" halves of the experiment exhibited comparable outcomes, it was further assumed that: *H3 : There is no significant two-way or three-way interaction with respect to the Modality factor.* This hypothesis was considered because, if there was interaction, then one experimental condition would have behaved differently than the other when combined with the other factors' levels.

Only general effects of human depth judgments were relevant to this experiment. Hence, no specific participant requirements were considered for the sample design. Overall, 32 participants were recruited within the extended institute of the authors for this study, nine of whom identified as female. They were between 23 and 46 years old (median 29 years) and between 164 cm and 205 cm tall (median 177.5 cm). 18 participants reported no visual impairments while the remaining 14 subjects had corrected-to-normal vision.

3.1 Tasks & Variables

Participants were instructed to repeatedly judge the relative depth order of five virtual cubes placed beneath a table that served as the projection surface. Vertical depth lines were provided to enhance spatial perception of the generated volume. Participants sorted the cubes along the height of the table, arranging them in a downward direction from top to bottom. The cubes were randomly labeled with the characters "P" through "T," with the labels projected onto the table surface. The table was 60 cm × 70 cm × 140 cm in size (width×height×depth) and covered with a white sheet. Cubes were placed randomly in predefined sub-cells within a 42 cm × 47 cm × 81 cm volume to avoid occlusion with one another. To this end, the specified volume was sub-divided in $5 \times 5 \times 5$ evenly spaced cells. No cell index in either direction was used more than once to avoid overlap and minimize additional depth cues. Depending on their distance, their edge length was between 6 cm and 11 cm with a random bias of ± 0.5 cm. More far away cubes had a larger scale factor to neglect effects of relative size depth cues.

Four dependent variables were measured during each sorting task. First, two efficiency measures were recorded: Task completion time (TCT) and total head movement (THM). Recordings started after a new set of sorting objects were presented and stopped after the experimenter noted the sorting answer. The correctness of this answer was assessed by the means of inversion numbers. This was calculated in R by comparing the ground truth sorting permutation

with the given answer permutation. The resulting inversion number was considered as the downward sorting error (DSE). Finally, the single ease question (SEQ) proposed by Sauro and Dumas [47] was used to assess the subjectively perceived task difficulty on a 7-point Likert rating scale.

3.2 Procedure

The experiment was conducted in a computer laboratory with constant lighting conditions ensured by ceiling lights and window blinds. After participants were welcomed, informed consent was collected and they were informed about the study, its procedure and data handling protocol. All data were collected anonymously, handled in compliance with applicable data protection regulations, and the study protocol conformed to the ethical principles of the Declaration of Helsinki. Afterwards, demographic information was gathered including age, sex, body height, and known vision deficiencies. Interocular distance was measured to calculate binocular disparity. The participants were then positioned in front of the table used as projection area. The subjects were instructed that they may move around freely in *front* of the table but may *not* move to its sides. After this setup, a training phase was conducted with the first device of the *Modality* factor. During this phase, the specific device was explained and equipped, its interocular distance settings were adjusted to the participant and the three visualization types were explained together with the general task. Then the first study trial commenced with the experimenter pressing a button. A new set of cubes appeared and the participants could start assessing the relative depth. They dictated the sorting order and the experimenter took notes and repeated the answer they heard. In case everything was correct, they pressed a button again and the cubes disappeared. The participants were then asked the SEQ "Overall, how easy or difficult did you find this task". After collecting this answer, the experimenter pressed the button again and the next trial commenced. This was repeated six times with two repetitions for each *Visualization* level. Then the *Disparity* mode switched and six additional trials were conducted in the same *Visualization* order. Afterwards the *Modality* device was exchanged and the whole process was repeated an additional time. Overall, this resulted in 24 measured trials. The order of devices was alternated every two participants while the order of *Disparity* modes was switched every participant. All six possible *Visualization* orders were precomputed and repeated every six participants. The study was concluded with two closing questions answered on a five-point Likert item together with an explanation: "How similar or different have you perceived the [monoscopic | stereoscopic] projections with and without the video see-through HMD?" ([|] indicates difference between both questions).

3.3 Statistical Analyses

All statistical analyses were conducted using R (version 4.4.0) in RStudio. First, mean values for repeated trials of identical experimental conditions were calculated (medians were used for SEQ data instead). No outliers were removed. Then, three-way repeated measures ANOVAs were calculated on the TCT, THM and DSE data. Greenhouse-Geisser corrections were applied in case of sphericity

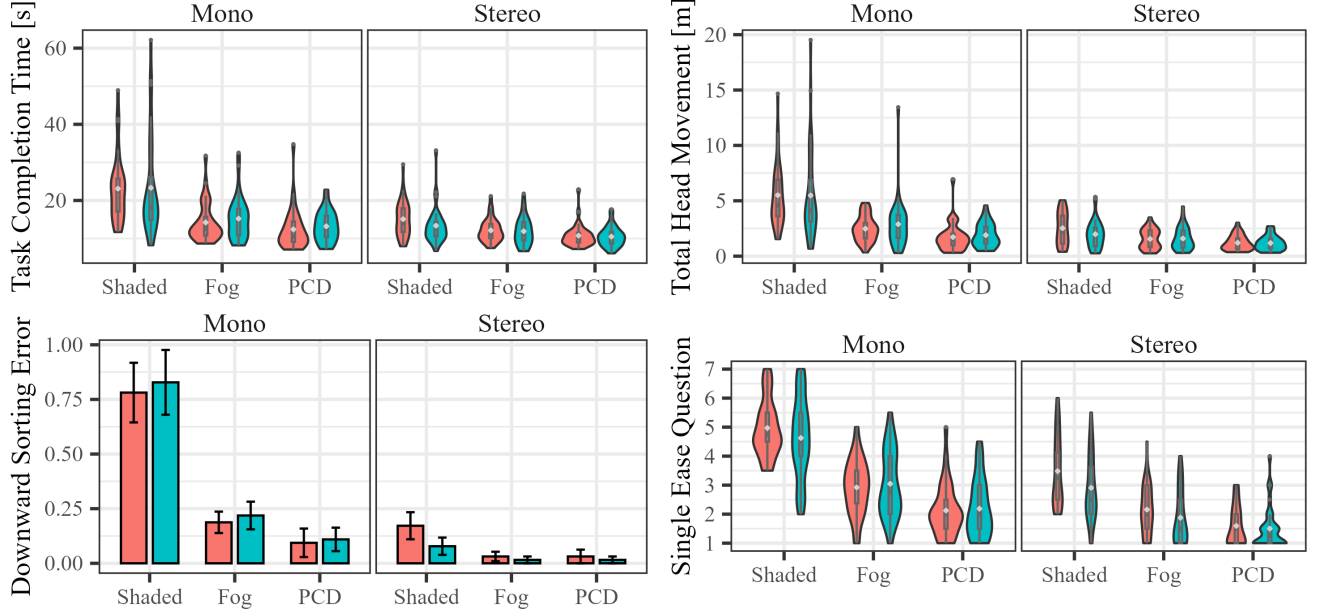


Figure 3: ■ - Projector-based AR, ■ - Video see-through AR. Experimental results. Violin and box plots visualize the distribution of the data for each experimental condition. Bar charts are shown for DSE data to enable better visibility of effects.

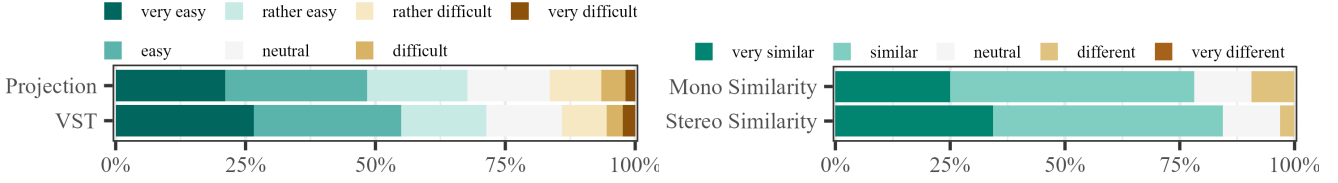


Figure 4: Distributions of SEQ scores between both AR devices and subjective similarity scores.

violations identified by Mauchly tests. Aligned rank transformations were applied to the ordinal scaled SEQ data using the *ARTool* package. Type III Wald F tests were then conducted on this data to assess main and interaction effects, while the participant ID was incorporated as a random effect factor. To further evaluate findings on the *Modality* factor, Bayes factors between *Projector* and *VST* variants for each combination of *Visualization* and *Disparity* levels were calculated. This was done to estimate how well the null hypotheses, that there are no differences between groups, could be supported [21]. Results are reported in Table 1 and interpreted according to the literature [30].

4 Results

Regarding TCT, statistically significant *Visualization* ($F(2, 62) = 39.68, p < 0.001$) and *Disparity* ($F(1, 31) = 48.39, p < 0.001$) main effects were identified. However, these are challenged by a significant *Visualization X Disparity* two-way interaction effect ($F(2, 62) = 11.28, p < 0.001$). Moreover, no statistically significant main effect on the *Modality* factor could be found ($F(1, 31) = 0.00, p = 0.95$). Likewise, the two-way interactions between *Modality* and *Visualization* ($F(2, 62) = 0.98, p = 0.36$), and between *Modality* and *Disparity*

Table 1: Summary of Bayes factor analyses results for each *Visualization* and *Disparity* combination between *Modality* levels. For each measure, BF_{10} is listed. Color Legend: ■ Strong evidence for H_0 , ■ Anecdotal evidence for H_0 , ■ Anecdotal evidence for H_a , ■ Moderate evidence for H_a .

Visualization	Disparity	TCT	THM	DSE	SEQ
Shaded	Mono	0.19	0.19	0.20	0.75
Shaded	Stereo	0.63	1.28	1.06	14.68
Fog	Mono	0.28	0.34	0.21	0.27
Fog	Stereo	0.21	0.19	0.22	2.74
PCD	Mono	0.25	0.24	0.21	0.21
PCD	Stereo	0.21	0.19	0.20	0.26

($F(1, 31) = 2.05, p = 0.16$) were not significant. In addition, no significant three-way interaction occurred ($F(2, 62) = 0.11, p = 0.89$). The data distribution is visualized in Figure 3 top left.

Similarly to TCT, the THM variable also showed statistically significant *Visualization* ($F(2, 64) = 45.32, p < 0.001$) and *Disparity* ($F(1, 31) = 46.47, p < 0.001$) main effects, as well as significant *Visualization X Disparity* two-way interaction ($F(2, 62) = 18.42, p <$

0.001). Again, no statistically significant main effect on the *Modality* factor could be found ($F(1, 31) = 0.00, p = 0.98$), and the two-way interactions between *Modality* and *Visualization* ($F(2, 62) = 2.48, p = 0.11$), and between *Modality* and *Disparity* ($F(1, 31) = 1.86, p = 0.18$) were not significant. The same was true for the three-way interaction ($F(2, 62) = 0.17, p = 0.84$). The THM data is further illustrated in Figure 3 top right.

Consistent findings also showed for the DSE measure. Statistically significant *Visualization* ($F(2, 62) = 28.31, p < 0.001$) and *Disparity* ($F(1, 31) = 27.46, p < 0.001$) main effects were identified, which were also challenged by a significant *Visualization X Disparity* two-way interaction effect ($F(2, 62) = 21.59, p < 0.001$). The *Modality* factor showed no significant differences ($F(1, 31) = 0.04, p = 0.85$) and its respective two-way interactions with the *Visualization* factor ($F(2, 62) = 0.10, p = 0.86$) and the *Disparity* factor ($F(1, 31) = 1.58, p = 0.22$) were also not significant. Moreover, no significant three-way interaction occurred ($F(2, 62) = 0.32, p = 0.73$). DSE data is visualized in Figure 3 bottom left. Bar charts are shown because of a floor effect caused by most data points being close to 0, making it difficult to interpret the resulting violin plots.

Statistically significant *Visualization* ($F(2) = 260.16, p < 0.001$) and *Disparity* ($F(1) = 202.14, p < 0.001$) main effects were also identified on the SEQ data. The two-way interaction between these factors was, again, significant ($F(2) = 17.37, p < 0.001$). In contrast to the other three variables, a statistically significant *Modality* main effect was found ($F(1) = 7.82, p = 0.01$). However, the respective two-way interactions with *Visualization* ($F(2) = 1.43, p = 0.24$) and with *Modality* ($F(1) = 2.12, p = 0.15$) did not show significance and also no significant three-way interaction was identified ($F(2) = 0.09, p = 0.91$). The general data distribution of the SEQ measure is illustrated in Figure 3 bottom right. In addition, Figure 4 left visualizes the distribution of collected SEQ data for the *Modality* factor to further visualize the observed significant main effect.

Bayes factors reported in Table 1 provide additional insights on how to interpret similarities or differences between the Projection and VST conditions of the *Modality* factor. Overall, there is anecdotal to moderate evidence supporting the null hypothesis that identical *Visualization* and *Disparity* combinations show no differences in means between the two *Modality* levels across dependent measures. However, there is also evidence supporting the respective alternative hypothesis that both *Modality* levels are different regarding the *Shaded Stereo* condition for THM, DSE and SEQ, as well as regarding the *Fog Stereo* condition for SEQ. Finally, Figure 4 right shows the results obtained from the concluding similarity questions. The majority of participants rated the monoscopic projections (78%) and the stereoscopic projections (84%) seen in the VST AR HMD and in the real environment as at least similar.

5 Discussion & Conclusion

Across all dependent variables, significant *Visualization* and *Disparity* main effects, as well as significant interaction between the two factors was shown. In the mean, the *Shaded* concept performed worse than the more depth-enhancing *Fog* and *PCD* techniques, with the latter showing best results. Concurrently, the *Stereo Disparity* mode performed significantly better than the *Mono* mode. However, an interaction effect occurred because stereoscopy had

a greater effect for the *Shaded Visualization* than for the other two techniques because they already conveyed sufficient depth information. This finding is in line with prior research, where similar effects could be shown [18], and confirms our hypothesis *H1*.

More noteworthy, the three-way statistical analyses did not reveal any three-way or two-way interaction effects with the *Modality* factor. This indicates that the effects explained above were not affected by the virtual content being shown in physical projector-based AR or in virtual projector-based-AR simulated in VST AR. Instead, they could be observed regardless of the *Modality*. This new finding confirms this work's hypothesis *H3*.

Regarding specific differences between both *Modality* levels, no significant differences were found for TCT, THM, and DSE. Pairwise Bayes factor analyses confirmed that there is in fact more evidence supporting the null hypothesis, that both *Modality* conditions showed comparable results, than the opposite. However, a significant main effect on SEQ indicates differences. The answer distributions seen in Figure 4 left shows that the VST simulation was generally rated to be easier than the physical *Projection*. Some participants remarked that they perceived the virtual projections to be sharper or clearer in color. However, these statements are challenged by other participants stating the reverse. Overall, the differences in the subjective assessment needs more investigations in future work and is not reflected by the objective data. Therefore, *H2* can at least partially be accepted. In addition, the vast majority of subjects rated both the monoscopic projections, and the stereoscopic projections to be similar in the concluding questionnaire. When asked for reasoning, most users referred their answers to general perceptual properties of VST AR, e.g., that the real world is only seen through the camera feed and is thus blurry, less sharp or otherwise altered. Some participants added that the fields of view were not identical which was either due to the VST HMD or due to the stereo shutter glasses. In addition, participants were in disagreement whether they perceived the physical or the virtual monoscopic projections to be more spatial. When asked to focus specifically on the projection, participants said to perceive close to no differences.

In conclusion, the study succeeded in showing that relative depth judgments are comparable between real and simulated projector-based AR visualizations. This is true for both monoscopic and stereoscopic displays and has two practical implications: On the one hand, this result is a first indication that other perception-related effects observed for projector-based AR are also applicable to virtual projection simulations. On the other hand, future findings determined using only VST- or VR-based simulations may also be likely to be applicable to real projection systems. This can facilitate future development processes and evaluations. However, the current findings should only be cautiously applied to future studies. More replication work, also proving that projection simulations are perceived comparably, is needed, to consolidate such systems as feasible options for physical projector-based AR.

If this can be consistently shown, virtual projections may be used whenever and wherever physical projector systems are reaching their limits. This could for example be due to economic or logistical shortcomings or because of temporal restrictions. However, in the end, projector-based AR simulations should not be used to

replace physical setups, but to complement them and facilitate their development process.

Limitations of this work include the general selection of the replicated effect. Only a relative depth sorting task was considered and no conclusions regarding absolute distance assessments can be made. In addition, the observed objects were only abstract and rather sparsely distributed in the volume. More complex and extensive visualizations, e.g., anatomical models, may result in different outcomes. Similarly, only three specific visualization concepts were considered in this work. Other studies also included methods like depth of field [48] or auxiliary supporting lines [18]. It remains ambiguous whether such techniques are also perceived similarly between both *Modality* methods evaluated in this work. Current findings are also only applicable to one specific VST device. Using the Apple Vision Pro (Apple Inc., USA) or even optical see-through devices like the Microsoft HoloLens (Microsoft Corporation, USA) could lead to other outcomes. Future studies are, thus, needed to investigate other devices, visualizations concepts, virtual contents and perception tasks.

In addition, differences in participant height and stereo vision acuity were not accounted for in this study. Regarding the former, height was recorded as a control measure but was not further used in subsequent analyses because this would have gone beyond the scope of this work. With respect to stereo acuity, only self-reported responses were collected but differences between participants may have still existed and could have impacted performance across conditions. Future work should account for both user-specific factors.

Another limitation is the chosen position and size of the projection area used in this work. The previous similar experiment of Heinrich et al. [18] used a much wider table as projection surface which potentially resulted in less occlusion depth cues between virtual objects. However, the physical projection setup available in this study could not provide identical surface dimensions. Nonetheless, similar findings could be reproduced. Thus, the effect of a smaller surface is believed to be marginal. On a similar note, future work could also investigate effects of differently oriented projection surfaces, e.g., walls. A planar surface was chosen in this work to better simulate projector-based AR applications displaying objects positioned below projection surfaces, e.g., for medical applications. However, investigating different orientations could result in meaningful contributions.

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